FitBuddy – Requirements Analysis

Project Name: FitBuddy - AI Fitness Plan Generator using Gemini Models

Detailed Project Documentation

1. Introduction

Requirements analysis identifies what the FitBuddy system must do, what information it needs and how users interact with the application.

Tabulation

Item	Details
Project	FitBuddy
AI Technology	Gemini Models
Type	Generative AI application
Purpose	Personalized wellness/activity planning

Coding / Technical Example

```
# FitBuddy technical flow
user_input = {
    "experience": "beginner",
    "activity": "walking",
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
# Display the generated response in the UI.
```

Picture / Diagram

```
IDEA → USER NEED → AI SOLUTION → FITBUDDY
```

2. Functional Requirements

The application should collect user details, validate inputs, construct prompts, communicate with Gemini, receive AI output, display a structured plan, allow regeneration and optionally save plan history.

Tabulation

Requirement	Description
Functional	Collect input, generate and display plan
AI	Gemini API integration
Security	Protect API credentials
Usability	Simple and clear interface

Coding / Technical Example

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    "experience": "beginner",
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```

```
    "time": "30 minutes"
}

prompt = f"Create a general wellness plan for {user_input}"
# Send prompt to Gemini from the secure backend.
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```

Picture / Diagram

```
USER REQUIREMENTS → FUNCTIONAL + NON-FUNCTIONAL → SYSTEM REQUIREMENTS
```

3. User Input Requirements

Typical inputs include fitness experience, preferred activity, available time, preferred number of days, equipment availability and general wellness goal.

Tabulation

Layer	Main Responsibility
Frontend	Collect and display information
Backend	Validation and application logic
AI Layer	Gemini prompt and generation
Database	Optional plan storage

Coding / Technical Example

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Picture / Diagram

```
FRONTEND → BACKEND → GEMINI API → RESPONSE → UI
```